Brighter Prospects?
Assessing the Franchise Advantage using Census Data*

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Abstract

This paper uses Census micro data to examine how starting a business as a franchise rather than an independent business affects its survival and growth prospects. We first consider the factors that influence the business owner's decision about being franchised, and then use different empirical approaches to correct for selection bias in our performance analyses. We find that franchised businesses on average benefit from higher survival rates and faster initial growth relative to independent businesses. However, the effects are not large and, conditional on first-year survival, the differences basically disappear. We briefly discuss potential mechanisms to explain these results.

Keyword: Independent Business, Business Survival, Business Growth, Franchising Decision, Survey of Business Owners, Retail, Services

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1. INTRODUCTION

Franchising is an important form of business organization in many retail and service industries, which themselves constitute a growing segment of the US economy. According to the U.S. Census Bureau, there were 453,326 establishments of franchised chains in 2007, contributing $154 billion in payroll for their 7.9 million employees. Given its importance in the U.S. economy, it is perhaps surprising that research on franchising decisions and the survival and growth prospects of franchising businesses remains scarce. The goal of this paper is to investigate whether franchised businesses' survival and growth prospects compare favorably to those of independent businesses.

The question of whether franchised businesses are more likely to be successful, and in particular whether they are more likely to survive longer and grow faster, is of much interest to industry participants as well as scholars. One regularly finds claims in the trade press suggesting that starting a franchise is a much safer way of going into business. The empirical evidence on this topic, however, has been mixed.[1] Bates (1995a, 1995b), for example, documented that in the late 1980s, the failure rate of franchised businesses was somewhat greater than that of independent businesses, and much greater than that suggested by industry insiders and the trade press.[2] More generally, in summarizing the literature on this topic, Blair and Lafontaine (2005) state:

> The data we present below show that substantial business risks exist for individuals who choose to invest in a franchise. In fact, franchising is no safer on average than independent business ownership, and in some cases is actually more risky. […] Of course, business failure is a fact of life. Our point is that this is just as true for franchisors and franchisees as it is for other players in the economy. We see this as no cause for alarm, but as a fact that should be more widely known and recognized in the franchise community. Ultimately, a false sense of security can only lead to hardship and disappointment.

In theory, it is indeed unclear whether franchised businesses or independent businesses

should have a survival advantage. On the one hand, starting a business as a franchise should be less risky than launching an independent business because franchisees benefit from their franchisor's brand name awareness and know-how and may realize cost savings from more efficient supply chains and bulk purchasing. That franchised establishments have become a ubiquitous part of modern life today indeed suggests that business owners value these benefits and that franchising is often an efficient mode of business entry and operation. Independent business owners, however, do not have to pay franchising fees, in particular ongoing royalties on revenues, and they retain complete autonomy, allowing them to adapt as needed to their local market and to changing market conditions. These differences allow for the possibility of superior performance by non-franchised businesses. The coexistence of both types of businesses in the marketplace further suggests that neither form of business ownership clearly dominates the other.

In this paper, we empirically revisit the question of whether the survival and growth prospects of franchising businesses are more promising than those of independent businesses using Census data. As noted in the literature, however, this empirical analysis requires caution because the decision to open a franchised business rather than an independent business is not random. The factors that affect the franchising decision may also affect survival directly. Thus, one must find a way to control for "selection into franchising" when examining whether franchising has a direct effect on survival (see, for example, Kosová, Lafontaine et al. (2013) on this issue). Evidence that selection into franchising by small business owners is not random is provided notably by Williams (1999), Kaufmann (1999), and Mazzeo (2004). Williams (1999), for example, shows that the characteristics of business owners, such as education, managerial experience, and experience as a salaried worker, increase the likelihood of entering into a franchise relationship relative to independent business startup. Our findings below on the factors that affect the probability of

starting a franchise relative to an independent business are in line with the findings in this literature.

Our paper also contributes to the broad literature on the determinants of survival and growth prospects of business startups.[3] The factors that have been identified in this literature can be grouped into three broad categories. The first category comprises characteristics of business owners, including race and gender, and measures of their human capital, such as education and an owner's prior business experience (e.g., Bates 1990, Boden and Nucci 2000, Cooper, Gimeno-Gascon et al. 1994, Cressy 1996, Evans and Leighton 1989, Fairlie and Robb 2008, Kalleberg and Leicht 1991, Kalnins and Mayer 2004, Lafontaine and Shaw 2015, Lazear 2004, Millán, Congregado et al. 2010, Pfeiffer 2000). The second category of factors relates to financial capital, usually referring to the amount of financial capital and the structure of financial sources that a business owner is able to garner to start their business (e.g., Bates 1990, Carpenter and Petersen 2002, Cooper, Gimeno-Gascon et al. 1994, Cressy 1996, Holtz-Eakin, Joulfaian et al. 1994, Schafer and Talavera 2007). The third category captures various aspects of the macroeconomic conditions faced by the business, including factors affecting the industry within which the new business operates, the local business environment, and the national level of economic activity (e.g., Agarwal and Audretsch 2001, Audretsch et al. 1995, 2000, Boden and Nucci 2000, Fritsch, et al. 2006, Haapanen and Tervo 2009, Pfeiffer and Reize 2000, Tveterås and Eide 2000, Wagner 1994). Note that a number of the factors in these three categories arise from decisions made by business owners. Examples include the form of financing, the timing of entry, the location and choice of business activities, and the skills that the entrepreneur chooses to acquire (see e.g., Cressy 1996, Kalnins and Lafontaine 2013, Lazear 2004, Pfeiffer and Reize 2000). The issue of selection mentioned above applies to these as well, and is addressed in some of the literature above. For our purposes, however, the main point is to recognize that these factors need to be controlled for in

our analyses of the effect of the franchising decision on both business survival and growth.

Our data are from the U.S. Census Bureau's Survey of Business Owners (SBO) and Longitudinal Business Database (LBD). The SBO provides the only comprehensive collected source of information on selected economic and demographic characteristics for businesses and business owners. The LBD includes information on all business formation and growth. We rely on the LBD to determine survival status for the first few years in business. To avoid effects arising from differences in survival rates documented for repeat business owners (see Bates (1998), Kalnins and Lafontaine (2013) and Lafontaine and Shaw (2016)), we limit our data to single-establishment businesses only. The industries covered in this study are the retail trade sector (NAICS 44-45),[4] the accommodation and food services sector (NAICS 72), and other services sector (NAICS 81). These sectors accounted for 43.4 percent, 27.6 percent, and 6.3 percent of franchisee businesses, respectively (Mesenbourg 2010).

We find that in simple mean comparisons, the one-year survival rate of new single-establishment businesses is about 6.3 percentage points higher for franchised than for independent businesses. The difference in the two-year survival rate is 8.4 percentage points. Controlling for factors that may lead an individual to pursue a franchise rather than open an independent business reduces the difference in survival rate slightly, to about 5 percentage points for the one-year survival rate, and 6 percentage points for the two-year survival rate. We employ Rosenbaum sensitivity tests as well as an instrumental variable (IV) and a control function (CF) approach to provide additional robustness tests. Overall, we find some evidence that franchised businesses have a better survival rate and faster growth than independent businesses in our data. Specifically, they benefit from greater odds of surviving in their first year that manifests itself in greater probability of also making it to two and three years. *Conditional on surviving for a year*, however,

the differences in survival or growth basically disappear. The absence of further survival or growth advantage conditional on the first year in business suggests that whatever the franchise model provides to business owners affects the chances they can make it through the crucial first year, but independent businesses that also survive their first year are more or less on an equal footing from that point on. We conclude that this initial franchise advantage is potentially explained by franchisor screening, which the market performs for independent businesses during the first year, or by franchisor-provided business know-how and consumer brand awareness advantages, advantages that the independent business owner can match – and/or that the franchisee fully pays for in the form of ongoing royalty and advertising fees – after the first year in business.

The paper is organized as follows. In the next section, we describe the data. In section 3, we present our empirical approach and results. We conclude in Section 4.

2. DATA

Whether a business is franchised or independent is identified in the Survey of Business Owners (SBO), a confidential U.S. Census Bureau survey conducted for the first time in 2002.[5] The Census's attempt to obtain information on more than 2.2 million businesses (and their owners) yielded a response rate of 75 percent in 2002 and 62 percent in 2007. As of the time of this writing, data collected in the 2012 survey have not yet been released.

The SBO universe includes all nonfarm businesses with annual receipts of at least $1,000 that filed Internal Revenue Service tax forms as individual proprietorships, partnerships, or corporations. The SBO collects information on the characteristics of both the business (e.g., sector of operation, legal form of organization, and franchise status) and the business owners (e.g., gender, race, age, and education). These variables enable us not only to examine the effect of

owner characteristics on business performance directly, but importantly, to control for differences between franchised and independent businesses and differences in the characteristics of their owners as well. As described further in Section 3 below, because many of the characteristics that affect the decision to buy a franchise or start an independent business also are expected to affect performance, were we not to control for them in our analyses, our estimates of the effect of franchising on survival or growth would likely be biased upward. In other words, our estimates could suggest that survival or growth is positively affected by franchising when in reality, the survival or growth differential may be attributable to differences in the characteristics of the businesses or their owners.

The Longitudinal Business Database (LBD) is another confidential Census Bureau database that provides annual information on all private sector employer establishments in the United States, i.e. all establishments with a payroll.[6] Merging the LBD with the SBO based on unique business establishment identifiers enables us to measure the survival of employer businesses surveyed in the SBO (the match is quite good, at 98.2 percent for 2002, and 98.8 percent for 2007).

To avoid effects arising from differences in survival rates documented for businesses started by owners with prior experience in franchising (see Bates (1998)) or in business more generally (see e.g. Kalnins and Lafontaine (2013) and Lafontaine and Shaw (2016)), we limit our data to single-establishment businesses only. To avoid survivorship bias in the data, namely cases where, in order to appear in the surveys, firms must have survived up to the survey year, we focus on businesses started in the years to which the SBO relates, that is, 2002 and 2007.[7] These restrictions reduce the sample size to 158,600 businesses in the 2002, and 96,700 businesses in the 2007, SBO.[8]

We further limit our analyses to privately-owned businesses whose main owner had at least a fifty percent stake. This restriction ensures that the business owner whose characteristics we control for indeed can influence the business and its success. To focus on for-profit enterprises, we also eliminate nonprofit establishments and those owned by an estate, a trust, or members of a club. Finally, our goal being to assess the success of franchised relative to independent businesses, we restrict our sample to those businesses operating in sectors where franchising is a real option. In particular, we include new businesses in the retail trade sector, in the accommodation and food services sector, and in other service sectors, which together account for about 77 percent of franchise businesses according to the Census Bureau. Within these broad sectors, we further limit our analyses to sub-sectors with a non-negligible franchising presence. We define this subset as four-digit NAICS industries in which at least three percent of businesses in our SBO data are franchised. Finally, we exclude automobile dealers and gasoline stations as regulation governs the franchise decisions in these sectors (in all states for automobile dealerships, and some states for gasoline stations). The final list of sectors is shown in Table 1. As one might have expected, limited-service eating places and traveler accommodations are the two sectors with the highest franchising rates.

Our final sample for the SBO 2002 and 2007 waves consists of 5,240 and 1,690 single-establishment businesses started in those two years.[9] Comparing with the 10.5 percent of businesses identified as franchised in the 2007 Economic Census, we find that 13.7 percent of businesses in our final combined sample are franchised. This relatively high fraction is, however, sensible given our focus is on franchising-intensive sectors.

For each business in our data, we generate a series of binary variables that indicate whether it is still operating *t* years after it was started. We run separate analyses for each survival duration

up to three years after business startup. It is important to note that the LBD records a business sold to another owner as an exit and entry of a new business. Because small business owners who sell their businesses often receive, according to the trade press, less than predicted future cash flows would warrant, business sales may well represent a type of failure, in which case our data correctly capture the event of interest.[10]

Table 2 reports descriptive statistics for all variables, by franchise status and SBO wave. Our main dependent variables, rates of business survival and employment growth, are higher for franchised startups. The one-year survival rate for the 2002 cohort is 98.3 percent for franchised, and 92.3 percent for independently owned, businesses. The two-year survival rates are 90.4 and 81.7 percent, respectively. The three-year survival advantage of franchised over independent businesses is even greater, at about 9.6 percentage points in 2002. This pattern holds also for the 2007 cohort, although all survival rates are smaller in that sample, likely reflecting the fact that businesses started in 2007 were soon faced with a major economic downturn. In terms of business growth, we use employment measures. We see that in the first year, both franchised and independent businesses have small employment numbers.[11] Starting from the second year, the numbers become larger and steadier.

Comparing business owner characteristics across new franchised and independent businesses, we see interesting evidence of "self-selection into franchising". About 70 percent of business owners are male in these data, and a greater percentage of franchised compared to independent businesses are started by men. People in the 25-34 age range are more likely to start an independent business, while 35-44 year olds are more likely to start a franchised business. Also, there is a difference in formal education for owners of the two types of businesses. People with a high school education or less, or with a vocational or technical education, are more likely to start

an independent business, while those with some college or more education are more likely to start a franchise.

When asked to describe how much they work, on average, independent business owners report working longer hours than franchise business owners. Franchised businesses are also more likely to be organized as corporations, and less likely to be operated from home. They also tend to be somewhat larger in terms of initial employment size, and more likely to rely on bank loans as a source of financing. In terms of location, franchise businesses are more likely to be located in counties in central Metropolitan Statistical Area (MSA) than independent businesses are.

To get a better understanding of the survival rates of retail and service businesses across industries in our data and across U.S. states, we summarize the survival prospects at both the sector and the state level. Ideally, we would like to provide these summary statistics on the data we analyze below. However, due to the U.S. Census Bureau disclosure criteria on sample size, as a compromise, we report summary statistics on the survival prospects for all the firms started during 2003 to 2007 covered in the LBD. The benefit of using this alternative is the larger sample size, but the downside is that the covered businesses do not conform to our data selection process, where we restricted our sample to businesses that are single-establishment, majority owned, non-seasonal, non-temporary, etc. Thus, the summary statistics, while informative generally, are only suggestive for our data. Table 3 reports the average one-year, two-year, and three-year survival rates across 4-digit NAICS industries. The three-year survival rates range from 54 to 78 percent in these data, with businesses in the direct selling sector having the lowest, and shoe stores having the highest survival rate. On average, though, the one, two and three-year survival rates for the new establishments in the LBD are lower, at 87, 73, and 63 percent respectively, than the corresponding 93, 82, and 72 percent survival rates we observe in our data. Figure 1 presents a

U.S. heat map of state-level business three-year survival rates for the larger LBD sample. We see that the Midwest and Northeast regions have relatively higher survival rates while the Southeast exhibits relatively lower survival rates. The one-year and two-year survival rates display similar patterns.

3. ECONOMETRIC SPECIFICATION

3.1 The Franchising Decision

In this section, we focus on the factors that affect a new business owner's decision to start a franchised or an independent business. We relate the franchising decision to a number of characteristics of the owner and the business:

$$Franchise_{ikcs} = f(X_{ikcs}, M_{ic}, \eta_k, \delta_s, \varepsilon_{ikcs}),$$

where $Franchise_{ikcs}$ is equal to one if the business is franchised, and zero otherwise, and the subscript i identifies the business, k indexes the 4-digit NAICS code for the business, c denotes the county, and s denotes the state. The X vector includes owner and business characteristics. Specifically, we include indicators for owner age, gender, race, and education, and sources of financing. As Mazzeo (2004) finds that heterogeneity in the economic environment is correlated with the franchising decision, we also include market-level, in this case county, characteristics for the year in which the business was started, such as the mean personal income per capita and demographic characteristics, denoted by M_{ic}. Lastly, to account for various common but unobserved constant factors within U.S. states and within NAICS industries, we include a set of state and a set of industry fixed effects (4-digit NAICS), denoted above by δ_s and η_k respectively. The explanatory variables are described further in Appendix A.

To assess the robustness of our results to functional form assumptions, we estimate the

franchise decision regression under a linear probability model, a probit model, and a logit model. We also report the results for three samples: the 2002, 2007, and the combined sample.

In the first three columns of Table 4 we show results from the linear probability model, with the probit and logit results in the next sets of columns. Though these are not shown in the table, we found that the marginal effects at the mean estimated from the two non-linear models are similar in terms of magnitudes and statistical significance levels to those we obtain from the linear probability model. Moreover, the results are also similar across samples. Consequently, in discussing results, we focus on those from the linear probability model for our combined sample.

Our results first confirm that the characteristics of owners do affect the decision to start a franchised business. For example, men are about 3 percent more likely than women, and Asians are 2.5 percent less likely than Caucasians (the default category), to start a franchised business. Consistent with the descriptive statistics in Table 2, education levels also matter. Entrepreneurs with higher education levels are more likely to start a franchised business: individuals with a bachelor degree are about 8 percent more likely than those with a high school degree or less to start a franchised business. In these analyses, we also control for sources of financing, which could be correlated with age and education and gender, and thus could bias our estimates of effects for these variables if we did not include them in the regressions. The results for these variables should be interpreted with caution, however, because the causality here could run the other way. We find that reliance on bank loans, government guaranteed loans,[12] and home equity loans is highly correlated with starting a franchised business, while credit card usage and statements of "no capital needed" are negatively correlated with starting a franchised business. Of course, if banks and government agencies believe that a franchised business is a safer way for entrepreneurs to start a business, or if franchising makes it easier for business owners to prepare their business plans and

provide extra information to financial institutions, then franchised business owners are more likely to obtain a loan from the government or financial institutions, and thus less likely to have to rely on credit cards.

Lastly, in these regressions, we control also for local market characteristics. We find that businesses in counties with more children and teenagers, larger populations, and lower unemployment rates are more likely to be franchised. But overall, very few of the local market demographics or characteristics matter for the business owner's decision.

3.2 Franchising Survival and Growth Prospects

3.2.1 Methods

The effect of franchising on the survival and growth prospects of businesses has long been a question of great interest to entrepreneurs, industry participants, and academics. Due to the "selection into franchising" problem, it remains a challenging question to address, however. With the aid of the U.S. Census Bureau restricted data access to the SBO and the LBD, we are better able to address this problem and thus obtain hopefully unbiased estimates of the effect of franchising.

When analyzing survival, the dependent variable is an indicator variable for whether the business is still in operation *t* years later. We also examine the effect of franchising and other factors on conditional survival, i.e. whether the business is still in operation at the end of year two (or three) *conditional on having survived the first (or first two) year(s)*. When analyzing growth, the dependent variable is the growth rate, defined as (current year value *minus* previous year value) / previous year value. [13] In general, we estimate the following equation:

$$Y_{ikcst} = f(Franchise_{ikcs}, X_{ikcs}, M_{ic}, \eta_k, \delta_s, \varepsilon_{ikcst}),$$

where, again, the subscript i identifies a business started in an SBO survey year (either 2002 or 2007), k indexes its sector (4-digit NAICS), c denotes the county, and s the state in which the business is located in. The subscript t on the dependent variable indicates the time frame over which we measure survival or growth. There are several advantages to using a linear probability model with dependent variables defined via indicator variables to model survival relative to using a duration model. First, it is easier to address the endogeneity or selection into franchising problem using different approaches such as propensity score matching, instrumental variables, or a control function. Second, with the selected approach, the analysis on survival, marginal survival, and growth are carried out under a single consistent framework, thereby facilitating interpretation and comparisons of results throughout. Third, our approach allows us to clearly distinguish a potential one-time franchise advantage from an ongoing conditional franchise advantage.

The main explanatory variable of interest in the regressions below is $Franchise$ status. The X vector again includes the set of business owner, business, and market characteristics described above. In addition, we include sets of indicator variables for business owner's decisions that could affect performance, namely the average number of hours worked in the business, the legal form of organization of the business, whether the business is operated primarily from home, and whether it is family owned.[14] We also control for the size of the business at startup by including the initial year employment level. We include all these variables, many of which represent owner decisions potentially correlated with the franchise status, so that we avoid bias in estimating the effect of franchising on survival and growth. Finally, we account for various common but unobserved factors within U.S. states and within business sectors by including state fixed effects and industry fixed effects (4-digit NAICS). Thus, essentially, the estimation exploits within-state and within-industry variation in survival and growth between franchised and

independent businesses, and controls for "selection into franchising" by including the many franchising decision determinants presented in Table 4.[15]

Although the detailed business owner, business, and market characteristics should do a decent job in controlling for the "selection into franchising" problem, there could remain bias caused by potential unobserved characteristics in the error term. We therefore further explore the effect of the franchising decision on business survival and growth prospects using other identification strategies, namely by relying on a propensity score matching approach, combined with Rosenbaum sensitivity tests, and then also by obtaining results based on both an instrumental variable (IV) approach and a control function (CF).[16]

3.2.2 Results

To fulfill the Census Bureau disclosure rules on sample sizes and to keep the discussion concise, in this section, we present only results that relate to our combined sample. In general, these are similar to the 2002 sample results, and consistent with those from the 2007 sample as well, except that the small size of the latter makes the results statistically less robust.

3.2.2.1 Survival and Growth Results with Controls for Potential Confounds

Results for the analysis of new single-establishment franchised and independent businesses one, two, and three-year survival rates using a linear probability model are reported in the first three columns of Table 5. Comparing with the mean differences in the one, two, and three-year survival rates between the franchised and independent businesses in the combined sample suggested in Table 2, which are 6.3 percent, 8.4 percent, and 9.1 percent, respectively, the results from the regression show some evidence of "selection into franchising". That is, after controlling for a rich set of business-owner, business, and market-level characteristics, the differences in the survival

rates between new single-establishment franchised and independent businesses become smaller, at 5.1 percent, 6.4 percent, and 6.4 percent for one, two, and three-year survival respectively.

The next set of columns in Table 5 shows the results of the effect of franchising on marginal survival rates, i.e., the 2-year survival rate conditional on survival for one year, and then the 3-year survival rate conditional on survival for two years. These results show that the marginal survival advantage for franchised businesses is smaller - about 2.1 percent for the second year and not statistically significantly different from zero for the third year survival conditional on two-year survival.

Finally, the last set of columns in Table 5 reports the results for employment growth. We see that from the first year to the second year, franchised businesses expand quickly, growing 48 percent faster than do independent businesses in terms of employment. However, after the second year, the growth rates are not different between the two organizational forms.

The noteworthy differences in survival in the summary statistics (Table 2) and the statistically significant results in Table 5 together provide some evidence that franchised businesses survive longer on average. This increased survival rate is a one-time persistent change, in that the 2-year and 3-year survival rates are also greater for franchised businesses. But conditional on having survived for a year or two, the difference in further survival rates between franchised and independent businesses is not statistically distinguishable from zero. The same is true for employment growth: franchised businesses grow much faster than independent businesses between their first and second year, but not in the years that follow.

One concern about the results in Table 5 is that the coefficient of the franchising indicator variable may be biased due to some remaining unobserved factors that might affect both the franchising decision and survival or growth. As mentioned earlier, we address this problem using

propensity score matching with Rosenbaum sensitivity tests in the next section, and then with IVs and a control function approach further below.

3.2.2.2 Propensity Score Matching Results

Matching is a common non-parametric technique to evaluate binary treatments in the absence of a randomized controlled experiment. The basic idea is to match each treated subject with one or several untreated subjects with very similar characteristics, and thereby form a "counterfactual" control outcome for the treated subject. With strong matching, the difference between the realized outcome for the treated and for the "counterfactual" is a consistent estimate for the treatment effect. Ideally, we would match each treated subject with a control subject with exactly the same characteristics on all the observables. However, this gives rise to the curse of dimensionality problem when a lot of covariates present. Rosenbaum and Rubin (1983) thus propose matching based on a propensity score, namely a measure of the probability of being treated. This measure summarizes the relevant characteristics and converts the multidimensional set of characteristics on which matching should be done to a single dimensional propensity score.

This approach requires that we first estimate a probit model in which the outcome variable is an indicator variable set to one if the business is franchised and zero otherwise. Since our goal is to match observations, the set of explanatory variables in this estimation contains those used in Table 4 plus all the owner decision indicator variables mentioned earlier.[17] Upon estimation of the propensity score, we use one-to-one no-replacement matching with a caliper radius of 0.25 on the common support of the treated and control groups. Then, we use balancing tests to assess whether the propensity score summarizes all the information in observables related to the treatment, so that once we condition on the propensity score, the observables provide no further information about the treatment status.

There are two main advantages to this approach in addressing the "selection into franchising" problem. First, unlike our analyses presented in the previous subsection, this method does not impose a linearity assumption. Second, this approach allows us to directly add into our analyses a variable that allows us to test how much a "hidden" variable could affect our findings.

We report the results in Table 6. The first two rows show the means of the treated sample as well as the matched control sample. The difference between the two groups is reported in the third row. In addition, the fourth and the fifth rows report the standard error and the t-statistics of the difference. We find that the magnitudes of the estimated effects of franchising on survival rates, marginal survival rates, and employment growth rates are similar to those in Table 5, suggesting that the linearity assumption that we relied on in Table 5 may be appropriate in this context.

The last two rows of Table 6 show results from the Rosenbaum bounds sensitivity tests (see Appendix B for details). These tests allow us to determine how likely it is that an unobserved variable, which we cannot control for in our probit estimation, could affect our conclusions regarding the effect of franchising on survival or growth. The critical cut-off value for Rosenbaum bounds (Rbounds) are defined as the highest value of the bound, referred to as e^{γ} in Appendix B, such that the measured treatment effect would still be statistically significant at the 5 percent level. In order to interpret the Rbounds cut-off values, we convert them into "maximum allowable deviations" reported in the last row. Specifically, given that the average probability of starting a franchised business is 13.7 percent, we calculate how large deviations an unobserved variable can maximally introduce on the probability of being franchised without undermining the statistically significant effect of franchising on survival or growth. Results in the first column then indicate that franchised businesses would be found to have a better chance of one-year survival if the

unobserved characteristic of the business or owner or local market that affects both the franchising decision and survival does not make the probability of starting a franchised business go outside the range of (6.2 percent, 27.6 percent). Compared to the mean probability of starting a franchised business of 13.7 percent, this allowable deviation range is quite generous. That is the unobserved factor would have to have substantial influence on the decision to franchise to affect our conclusion that the first-year survival of franchised businesses is greater than that of independent businesses. This analysis implies that the results are quite robust to the potential existence of "hidden bias." Overall, we conclude that the findings from the propensity score matching and the Rosenbaum bounds sensitivity tests confirm our result from Table 5 that new single-establishment franchised businesses have a greater chance of survival, and grow faster, than new single-establishment independent businesses, but that the effects are relatively small. Also, while franchising increases the probability of businesses surviving one, two and three years in business to a similar extent, this occurs because of the one-time shift in survival probabilities early on in the life of the new businesses.

We conclude that the franchise advantage constitutes a one-time benefit that allows franchise businesses to make it through their very first year or two with somewhat greater probability, an advantage in the range of five to seven percentage points for the one-year to three-year survival in Table 6. However, once the small business has been in operation for a year or two, there is no further contribution of franchising to its survival or growth. Given slightly larger first year employment levels in franchised business, the greater growth we observe between year one and two also implies that franchised establishments will be somewhat larger on average than non-franchised establishments in the same industry. Our descriptive statistics in Table 2 confirm that this is the case on average in the data as well.

3.2.2.3 Instrumental Variable and Control Function Approaches to Identification

In addition to the propensity score matching approach combined with sensitivity tests above, we also use both an instrumental variable (IV) approach and a control function (CF) approach to address the identification problem that is central to correctly assessing the effect of franchising on survival and growth (see Appendix C for implementation details). Because the CF approach is less robust than the IV approach, but more efficient if the model for the binary endogenous variable is correctly specified, in what follows we report and discuss results from both approaches.

For both of these identification strategies, it is important to identify variables that would affect the decision to franchise, but not the performance (survival or growth) of the establishments. To do so, we make use of several such variables. In particular, using data from the 2007 Economic Census which, for the first time, included questions on franchising, we calculate the state-level proportion of franchised businesses in the same industry (6-digit NAICS level) and the same proportion in related industries (all 6-digit NAICS codes in the same 2-digit NAICS code, excluding the one associated with the business in question). We believe that the proportion of establishments that are franchised in the focal or related industries will affect a business owner's decision to start a franchise because they will be positively correlated with the availability and popularity of the franchising organizational form in the relevant industries and state, which in turn also will affect the business owner's awareness of this option for his business. At the same time, these variables should not have a direct effect on the survival and growth of any particular establishment. Thus, the average franchise proportions meet the two properties (informative and exclusive) for a valid instrument. We also use the state-level proportion of company-owned establishments of franchised chains in the same industry (6-digit NAICS level) as another excluded (from the second stage) instrument.[18] The argument for this instrument is as follows: as franchise

chains choose to own more establishments instead of turning to franchising, fewer opportunities are available for potential entrepreneurs. Thus one expects that the state-level proportion of company-owned establishments in the industry should have a negative effect on the franchising decision. At the same time, there is no reason to expect that this proportion would have a direct effect on an individual business' survival or growth prospects.

The summary statistics for these instrumental variables are provided in Table D1 in Appendix D. The first step regression results, in Table D2, also in that Appendix, confirm our expectation that greater reliance on franchising in the industry or in related industries has a positive effect, and the proportion of company-owned establishments has a negative effect, on an entrepreneur's decision to start a franchised business.

We report results for new single-establishment business survival and growth using the IV and the CF approaches in Tables 7 and 8 respectively. Though the first stage results (see Appendix D) suggest that our analyses do not suffer from the weak instrument problem, the estimated standard errors on the estimated coefficients are very large, such that basically none of the survival results in Tables 7 or 8 are statistically significant. Results for employment growth from years one to two, however, continue to suggest sizable and significant positive effects of franchising.

In general, from the IV results, we see that after correcting for potential endogeneity, the estimated effect of franchising on business survival rates become smaller, suggesting again the existence of potential upward bias in Table 5. The estimated coefficients on the generalized residual from the CF approach (Table 8) provide a possible explanation as to why the results from Table 5 differ from those obtained with the IV and the CF approaches. Though generally not statistically different from zero, the coefficients on the generalized residual in the CF approach imply that potential unobserved factors that are likely to be positively correlated with the

franchising decision also have a positive effect on the survival rates and a negative effect on the employment growth rate. One unobserved variable that could explain such relationships is the unmeasured business owner's risk aversion. If an entrepreneur is very risk averse, she is more likely to start a franchised business, less likely to expand the business dramatically in the short run, and the owner is more likely to have garnered enough capital and to behave in such a way that the business is likely to survive longer. In this case, if we did not, as we do not in Table 5, control for the business owner's risk aversion, the results would represent an overestimation of the effect of the franchising decision.

While we believe that the results in Table 5 might still somewhat overestimate the effect of franchising, it is also clear from Tables 7 and 8 that our IV and control function results are very noisy. This, also, is not uncommon, and it suggests caution in interpreting this last set of results as well. We therefore conclude that the preponderance of evidence suggests that franchised businesses benefit from somewhat higher survival rate and faster growth relative to independent businesses. However, conditional on surviving one year, we find no evidence that franchised businesses benefit from further enhanced survival or growth prospects.

We would argue that there are two main sources of potential survival and growth advantages for franchised businesses: either these arise from selection of better opportunities by franchisors, i.e. selection based on fixed characteristics, such as higher ability owners or better locations. Or it arises from the transfer to the franchisee of the franchisor's know-how and business format, including supply networks and brand awareness, which gives the franchised business owner an advantage over the independent business owner at startup. If franchisors' selection or screening processes allow only better business opportunities to be started as franchises, the lack of further advantage survival or growth advantage beyond a year or two would imply that the market

successfully screens the businesses of independent owners within the first year or so. If franchisees survive better in the first year because of important knowledge that the franchisor conveys to them, then the lack of further advantage conditional on the first year or two would imply that owners of independent businesses that survive a full year or so are able to develop the needed expertise to compete successfully with franchised businesses rather rapidly as well. Of course, since franchisees must pay ongoing fees to obtain and maintain access to the expertise and resources provided by their franchisor, the long-term coexistence of both types of businesses in the marketplace is consistent with the conclusion that neither form dominates the other in a general sense, and thus with our finding that there is not an ongoing advantage of franchising once the firm has been in business for a year or two.

3.2.2.4 Other Factors Affecting Survival and Growth

In addition to analyzing the effects of the franchise decision on survival and growth, our analyses also shed light on other factors affecting survival and growth for the new single-establishment businesses in our data. In Table 5, we find that businesses of African American owners have lower survival rates, and businesses of Asian owners have higher survival rates, compared with Caucasian owners. Also, the gaps increase over time, with the one-year survival rate difference being -5.8 percent and 0.2 percent and the three-year survival rate difference being -13.1 percent and 4.5 percent, respectively. Business owners in the 45-54 age group perform best, with a three percent higher three-year survival rate compared with those who are under 35 years old. We also find that compared to businesses started by people who have a high-school education level or less, those started by people who have a technical or vocational education or a bachelor's degree have greater survival rates, especially for the two-year and three-year. However, businesses of owners with some college or a college degree, or an associate degree, or a postgraduate degree, have no

greater survival rates than businesses of owners who have less than or equal to high-school education level.

Turning to variables that are under the direct control of the business owner at the time they start their businesses or thereafter, we find, not surprisingly, that working hard pays off for business owners. Businesses of owners who work at least 40 hours per week have about seven percent higher chance of survival after three years compared with businesses of owners who work less than 20 hours per week. Of course, given that weekly working hours is a decision variable for the business owner, these results could mean that working hard really pays off through higher business survival rates, as per the interpretation above, or that owners put less effort into businesses that are not promising. Still, the observed correlation between effort and survival is worth noting. Similarly, we find that businesses organized as corporations, the default category, have substantially greater survival rates than those organized as proprietorships or partnerships, and businesses operated from home have substantially lower survival probabilities. These results again might be capturing the effect of business owners' level of dedication to their ventures, but in either case, the association of these characteristics with survival is of interest generally. Further, initial year employment has some positive effect on the two-year and three-year survival rates. Lastly, we see a high positive correlation between reliance on a government-guaranteed loan and survival, and a negative correlation between other sources of capital and the survival rate. The high correlation between the reliance on government-guaranteed loans and survival might be the result of business owners' higher effort incentives when they have more at stake (e.g., when their home is used as collateral, as in Fan, Kuhn, and Lafontaine, 2017) or the result of the effectiveness of government screening processes in selecting higher ability candidates.

4. CONCLUSION

In this paper, we used restricted-access micro data from the U.S. Census Bureau, namely the 2002 and 2007 Survey of Business Owners and the Longitudinal Business Database, to investigate why entrepreneurs become franchisees. We found that an entrepreneur's characteristics, including gender, race and education levels, matter for this decision. Only a few market-level characteristics, notably population levels and proportion of children in the population, appear to be important for this decision.

The goal of this paper, however, was to analyze whether franchising confers a survival and growth advantage to new businesses. To address the "selection into franchising" problem, we examined the effect of franchising on survival and growth in a regression framework, controlling directly for the many owner and business and market characteristics that are associated with the decision to start a franchised rather than an independent business. We then revisited the effect of franchising on survival and growth using propensity score matching combined with Rosenbaum bounds sensitivity tests. Using the above approaches, we find that franchising is a somewhat safer way to start a business. The magnitude of the effect ranges from five to seven percentage points depending on specification. However, conditional on surviving for a year or two, we find no further franchise advantage in terms of survival. Similarly, we find evidence that franchised businesses grow faster than independent businesses in the first two years, but no difference beyond that point.

We address the identification of the franchise effects problem even further by using both an instrumental variable and a control function approach. In these analyses, we rely on data collected as part of the 2007 Economic Census to measure the state-level proportion of establishments operating as part of franchised chains in the same industry and in related industries, and the state-level proportion of company-owned establishments of franchised chains in the same

industry, as excluded-from-the-second-stage instruments. Because the results using these approaches are not statistically significant, we view them as suggestive evidence that there may still be unobserved characteristics that could bias coefficients in our earlier analyses. For example, some characteristics of the business or the owner, such as the latter's (unmeasured) risk aversion, could be affecting the decision to franchise and the performance of franchised units in a way that leads to upward biases in our estimates of the effect of franchising on establishment performance (survival and growth). Nonetheless, the lack of significance may also result from the increased estimated standard errors after using IVs.

We therefore conclude that while we find some evidence of a survival and growth advantage of franchising, this advantage is rather small and concentrated in the first year or two. Moreover, even the small advantage we measure may be overestimated. One interpretation of these results is that franchisors screen potential franchise owners and select business characteristics, thereby preventing some less developed or valuable ideas from reaching the market. Independent businesses are subjected to this screening by the market in their first year in business instead, leading to more failures of independent businesses in the first year. Alternatively, but also plausibly, franchisees benefit from the business format and knowledge and brand awareness obtained by being part of a franchise rather than being on their own in their first year in business. But after the first year or two in business, the fees that franchisees pay for the services of their franchisors on an ongoing basis are such that they make the profitability of these businesses equivalent to that of independent businesses. We cannot distinguish these two interpretations in the data. Moreover, we expect both are at play given that both explanations are consistent also with the continued coexistence of both types of businesses in the marketplace.

Finally, our analyses also shed light on other determinants of business survival for these

small businesses and business owners. In particular, we find that some business owner characteristics, including race, age and education, impact the survival probabilities of new businesses. Not surprisingly, we also find that the business owner's effort level, measured by the weekly number of hours worked, pays off in terms of survival and growth. Similarly, the choice of legal form, namely the decision to incorporate the business rather than run it as a sole proprietorship or via a partnership is positively related to its survival and growth. Lastly, access to government-guaranteed loans is strongly positively associated with survival and short-term growth. Effort, legal form, and type of financing are all possibly determined jointly by the business owners based on factors that may also directly affect survival and growth. As a result, while informative, these effects cannot be interpreted in a causal way.

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Table 1: The Proportion of Franchised Businesses across Sectors

Sector	NAICS4	Franchised Business (%)	Number of Estab.
Automotive Parts, Accessories, and Tire Stores	4413	15.5	170
Furniture Stores	4421	7.8	120
Home Furnishings Stores	4422	6.9	180
Electronics and Appliance Stores	4431	8.7	240
Building Material and Supplies Dealers	4441	10.2	150
Lawn and Garden Equipment and Supplies Stores	4442	7.4	50
Grocery Stores	4451	9.0	370
Specialty Food Stores	4452	22.9	170
Health and Personal Care Stores	4461	8.4	250
Clothing Stores	4481	3.3	210
Shoe Stores	4482	ND	40
Jewelry, Luggage, and Leather Goods Stores	4483	3.3	60
Sporting Goods, Hobby, and Musical Instrument Stores	4511	9.2	260
Other General Merchandise Stores	4529	9.9	80
Office Supplies, Stationery, and Gift Stores	4532	8.1	270
Other Miscellaneous Store Retailers	4539	10.7	250
Electronic Shopping and Mail-Order Houses	4541	3.3	60
Direct Selling Establishments	4543	3.3	60
Traveler Accommodation	7211	34.5	190
Full-Service Restaurants	7221	8.7	950
Limited-Service Eating Places	7222	35.1	1000
Automotive Repair and Maintenance	8111	9.8	790
Electronic Equipment Repair and Maintenance	8112	3.5	90
Personal and Household Goods Repair and Maintenance	8114	7.2	100
Personal Care Services	8121	10.8	560
Dry cleaning and Laundry Services	8123	4.5	110
Other Personal Services	8129	11.6	160
Total		13.7	6,930

Notes: To satisfy U.S. Census Bureau disclosure procedures, we report the franchise proportion among new businesses in the combined 2002 and 2007 sample instead of each year and we suppress information for a few sectors that fail to meet the disclosure requirements, denoted as ND. All counts are rounded to tens also to satisfy these requirements. The total in the last column is not equal to the sum of the numbers in that column due to the required rounding.

Table 2: Summary Statistics

Means	Businesses Started in 2002			Businesses Started in 2007			Combined Sample			
Dependent Variables	**All**	**Franch.**	**Indep.**	**All**	**Franch.**	**Indep.**	**All**	**Franch.**	**Indep.**	**t-stat**
One Year Survival	0.93	0.98	0.92	0.91	0.97	0.90	0.93	0.98	0.92	6.88***
Two Year Survival	0.83	0.90	0.82	0.80	0.86	0.79	0.82	0.89	0.81	6.18***
Three Year Survival	0.73	0.81	0.72	0.71	0.77	0.70	0.72	0.80	0.71	5.46***
1st Year Employment	1.06	1.80	0.94	1.31	2.30	1.15	1.12	1.93	0.99	6.01***
2nd Year Employment	7.37	14.48	6.17	8.03	15.3	6.77	7.53	14.68	6.32	17.27***
3rd Year Employment	7.90	14.39	6.76	8.21	15.15	6.99	7.97	14.57	6.81	16.73***
4th Year Employment	8.19	14.52	7.05	8.80	15.65	7.59	8.34	14.79	7.18	15.10***
Owner-Level Variables										
Male	0.69	0.77	0.67	0.67	0.76	0.66	0.68	0.77	0.67	5.96***
Caucasian	0.84	0.82	0.84	0.81	0.85	0.81	0.83	0.83	0.83	-0.46
African American	ND	ND	ND	ND	ND	ND	0.02	0.02	0.02	0.63
Asian	0.13	0.15	0.13	0.15	0.13	0.16	0.14	0.14	0.14	0.50
Other Race	ND	ND	ND	ND	ND	ND	0.01	0.01	0.01	-0.51
Age: Younger than 25	ND	ND	ND	ND	ND	ND	0.02	0.02	0.02	-0.78
Age: 25-34	0.21	0.19	0.22	0.18	0.14	0.19	0.21	0.18	0.21	-2.41***
Age: 35-44	0.35	0.38	0.34	0.34	0.37	0.33	0.35	0.38	0.34	2.35***
Age: 45-54	0.28	0.30	0.28	0.31	0.33	0.31	0.29	0.31	0.29	1.32*
Age: 55-64	0.11	0.10	0.11	0.13	0.14	0.12	0.11	0.11	0.12	-0.81
Age: 65 and Older	ND	ND	ND	ND	ND	ND	0.02	0.01	0.03	-2.28**
Less than High School	ND	ND	ND	ND	ND	ND	0.04	0.02	0.05	-4.29***
High School	ND	ND	ND	ND	ND	ND	0.09	0.05	0.10	-4.91***
Technical or Vocational School	0.25	0.17	0.26	0.24	0.19	0.24	0.24	0.18	0.25	-5.07***
Some College	0.21	0.22	0.21	0.16	0.16	0.17	0.20	0.21	0.20	0.43
Associate Degree	0.06	0.07	0.06	0.09	0.07	0.09	0.07	0.07	0.07	0.57
Bachelor's Degree	0.26	0.35	0.25	0.28	0.39	0.27	0.27	0.36	0.25	6.93***
Postgraduate Degree	0.09	0.12	0.09	0.10	0.15	0.09	0.09	0.12	0.09	3.68***
Weekly Work: Less than 20 Hrs.	0.14	0.18	0.14	0.14	0.21	0.13	0.14	0.19	0.13	4.44***

Weekly Work: 20-39 Hrs.	0.12	0.12	0.12	0.13	0.19	0.12	0.12	0.13	0.12	1.04
Weekly Work: 40 Hrs.	0.11	0.10	0.12	0.14	0.08	0.15	0.12	0.09	0.12	-2.48***
Weekly Work: 41-59 Hrs.	0.29	0.26	0.29	0.28	0.23	0.29	0.29	0.25	0.29	-2.47***
Weekly Work: 60 Hrs. or More	0.34	0.34	0.34	0.31	0.30	0.31	0.33	0.33	0.33	0.12
Business-Level Variables										
Franchise	0.14	1	0	0.14	1	0	0.14	1	0	N/A
Sole Proprietorship	0.21	0.12	0.22	0.20	0.12	0.21	0.21	0.12	0.22	-6.57***
Partnership	0.20	0.22	0.20	0.28	0.33	0.28	0.22	0.25	0.22	1.80**
Corporation	0.59	0.66	0.58	0.52	0.55	0.52	0.58	0.63	0.57	4.46***
Operated from Home	ND	ND	ND	ND	ND	ND	0.08	0.05	0.09	-4.36***
Family Owned	0.41	0.51	0.40	0.37	0.53	0.34	0.40	0.51	0.38	7.54***
Initial Year Employment	1.25	2.03	1.12	1.51	2.74	1.31	1.31	2.21	1.17	5.92***
Financing - Savings	0.70	0.67	0.71	0.72	0.70	0.72	0.71	0.68	0.71	-2.12**
Financing - Bank Loan	0.31	0.46	0.29	0.28	0.42	0.25	0.30	0.45	0.28	10.99***
Financing - Other Personal Assets	0.23	0.22	0.23	0.16	0.17	0.16	0.21	0.21	0.22	-0.49
Financing - Credit Card	0.19	0.13	0.20	0.22	0.17	0.23	0.20	0.14	0.21	-4.66***
Financing - Government Loan	0.04	0.06	0.04	0.03	0.06	0.02	0.04	0.06	0.03	3.64***
Financing - Government Guaranteed Loan	0.06	0.13	0.05	0.05	0.10	0.04	0.06	0.12	0.05	9.83***
Financing - Outside Investor	0.06	0.07	0.06	0.33	0.39	0.32	0.13	0.14	0.12	1.72**
Financing - None Needed	ND	ND	ND	ND	ND	ND	0.03	0.01	0.03	-3.73***
Financing - Home Equity Loan	N/A	N/A	N/A	0.22	0.29	0.20	N/A	N/A	N/A	N/A
Financing - Loan from Family/Friends	N/A	N/A	N/A	0.07	0.06	0.07	N/A	N/A	N/A	N/A
Financing - Other Sources of Capital	N/A	N/A	N/A	0.04	0.04	0.04	N/A	N/A	N/A	N/A
Financing - Grants	N/A	N/A	N/A	ND	ND	ND	N/A	N/A	N/A	N/A
Financing - Venture Capital	N/A	N/A	N/A	ND	ND	ND	N/A	N/A	N/A	N/A
Financing - Do Not Know	N/A	N/A	N/A	ND	ND	ND	N/A	N/A	N/A	N/A
Market-Level Variables (Counties)										
Population	699802	685051	702138	750254	637617	768331	712120	673367	718277	-0.99
Personal Income (Current $)	31254	31410	31229	39746	39501	39785	33327	33403	33315	0.25
Males	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0
Caucasian	0.83	0.82	0.83	0.80	0.80	0.80	0.82	0.82	0.82	-1.01

African American	0.12	0.13	0.12	0.13	0.14	0.13	0.12	0.13	0.12	1.91**
Other Race	0.05	0.05	0.05	0.07	0.06	0.07	0.06	0.05	0.06	-1.62**
Age: 14 and Younger	0.21	0.21	0.21	0.20	0.20	0.20	0.21	0.21	0.21	4.58***
Age: 15-64	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.99
Age: 65 and Older	0.13	0.12	0.13	0.13	0.12	0.13	0.13	0.12	0.13	-4.77***
At Least High School Degree	0.87	0.87	0.87	0.86	0.88	0.86	0.87	0.87	0.87	2.70***
Unemployed	0.06	0.06	0.06	0.05	0.05	0.05	0.05	0.05	0.05	-1.91**
Geographic Area (Square Miles)	1203	1178	1208	1309	13229	1306	1229	1214	1232	-0.26
MSA Central County	0.66	0.71	0.65	0.69	0.74	0.69	0.67	0.72	0.66	3.59***
MSA Outside County	0.08	0.08	0.08	0.09	0.08	0.09	0.08	0.08	0.08	-0.42
Non-MSA County	0.26	0.21	0.27	0.22	0.18	0.23	0.25	0.20	0.26	3.63***
Observations	5,240	720	4,520	1,690	230	1,460	6,930	950	5,980	

Notes: The last column reports two-sided equal-variance t-tests between franchised and independent businesses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Variables unavailable in one of the SBO surveys are denoted by N/A and cells that failed the disclosure criteria are denoted as ND. *Source:* U.S. Census Bureau SBO 2002, SBO 2007, and LBD.

Table 3: Survival Rates across Sectors

Sector	NAICS4	Obs.	Survival Rates		
			1-year	2-year	3-year
Automotive Parts, Accessories, and Tire Stores	4413	20,330	0.89	0.79	0.71
Furniture Stores	4421	15,460	0.88	0.76	0.64
Home Furnishings Stores	4422	16,380	0.88	0.76	0.65
Electronics and Appliance Stores	4431	32,910	0.86	0.72	0.62
Building Material and Supplies Dealers	4441	22,560	0.89	0.79	0.71
Lawn and Garden Equipment and Supplies Stores	4442	6,680	0.89	0.79	0.70
Grocery Stores	4451	48,760	0.86	0.72	0.62
Specialty Food Stores	4452	21,510	0.84	0.68	0.56
Health and Personal Care Stores	4461	36,900	0.89	0.79	0.72
Clothing Stores	4481	50,500	0.90	0.78	0.69
Shoe Stores	4482	9,400	0.92	0.85	0.78
Jewelry, Luggage, and Leather Goods Stores	4483	11,620	0.87	0.71	0.63
Sporting Goods, Hobby, and Musical Instrument Stores	4511	21,710	0.88	0.75	0.64
Other General Merchandise Stores	4529	16,760	0.92	0.83	0.75
Office Supplies, Stationery, and Gift Stores	4532	21,710	0.86	0.72	0.61
Other Miscellaneous Store Retailers	4539	26,800	0.85	0.71	0.61
Electronic Shopping and Mail-Order Houses	4541	10,470	0.83	0.67	0.56
Direct Selling Establishments	4543	9,570	0.79	0.64	0.54
Traveler Accommodation	7211	26,020	0.89	0.79	0.71
Full-Service Restaurants	7221	134,950	0.87	0.71	0.60
Limited-Service Eating Places	7222	151,560	0.88	0.74	0.64
Automotive Repair and Maintenance	8111	75,770	0.84	0.71	0.61
Electronic Equipment Repair and Maintenance	8112	6,110	0.86	0.71	0.61
Personal and Household Goods Repair and Maintenance	8114	11,710	0.83	0.69	0.58
Personal Care Services	8121	73,210	0.85	0.71	0.61
Dry cleaning and Laundry Services	8123	21,450	0.87	0.74	0.64
Other Personal Services	8129	28,020	0.83	0.68	0.57
Total		928,830	0.87	0.73	0.63

Notes: All counts are rounded to tens to satisfy U.S. Census Bureau disclosure requirements. The data used in this table relates to all establishments started in any year between 2003-2007 (including 2003 and 2007), which is a much larger set of establishments than the one we focus on in our analyses below. We do this to have a larger data set to calculate survival rates, but of course this implies that the above data are not descriptive of the single-establishment, single owner, non-seasonal, non-temporary, etc. businesses that we focus on. In other words, the above information is only suggestive for our sample. And indeed, a comparison of the survival rates across the total sample shows that the businesses in our data have greater rates of survival than do the new businesses in the broader LBD.

Table 4: Determinants of the Franchise Decision

	Linear Probability Model			Probit Model			Logit Model		
VARIABLES	**2002**	**2007**	**2002 & 2007**	**2002**	**2007**	**2002 & 2007**	**2002**	**2007**	**2002 & 2007**
Male	0.027***	0.050***	0.032***	0.141**	0.281***	0.168***	0.269**	0.554***	0.329***
	(0.010)	(0.019)	(0.009)	(0.058)	(0.105)	(0.050)	(0.110)	(0.211)	(0.095)
African American	ND	ND	-0.001	ND	ND	-0.012	ND	ND	-0.021
	ND	ND	(0.030)	ND	ND	(0.147)	ND	ND	(0.270)
Asian	-0.0170	-0.032	-0.025**	-0.107	-0.176	-0.147**	-0.166	-0.296	-0.231**
	(0.014)	(0.022)	(0.012)	(0.074)	(0.133)	(0.064)	(0.135)	(0.254)	(0.116)
Other Race	ND	ND	-0.019	ND	ND	-0.141	ND	ND	-0.197
	ND	ND	(0.030)	ND	ND	(0.184)	ND	ND	(0.340)
Age: 35-44	0.024**	0.029	0.027***	0.141**	0.188	0.152***	0.245**	0.339	0.275**
	(0.012)	(0.022)	(0.010)	(0.064)	(0.124)	(0.057)	(0.122)	(0.243)	(0.107)
Age: 45-54	0.027**	0.018	0.027**	0.171**	0.141	0.164***	0.297**	0.263	0.293***
	(0.012)	(0.023)	(0.011)	(0.068)	(0.129)	(0.059)	(0.128)	(0.254)	(0.111)
Age: 55-64	0.001	0.043	0.015	0.001	0.232	0.077	-0.020	0.459	0.138
	(0.015)	(0.030)	(0.014)	(0.092)	(0.154)	(0.078)	(0.174)	(0.297)	(0.146)
Age: 65 and Older	ND	ND	-0.025	ND	ND	-0.143	ND	ND	-0.311
	ND	ND	(0.023)	ND	ND	(0.169)	ND	ND	(0.342)
Technical or Vocational School	ND	ND	-0.011	ND	ND	-0.112	ND	ND	-0.206
	ND	ND	(0.013)	ND	ND	(0.096)	ND	ND	(0.191)
Some College	0.046***	0.023	0.039***	0.270***	0.137	0.222***	0.523***	0.267	0.431***
	(0.013)	(0.025)	(0.011)	(0.073)	(0.134)	(0.064)	(0.139)	(0.266)	(0.121)
Associate Degree	0.080***	0.018	0.062***	0.429***	0.129	0.338***	0.843***	0.254	0.661***
	(0.020)	(0.031)	(0.017)	(0.104)	(0.174)	(0.088)	(0.193)	(0.343)	(0.164)
Bachelor's Degree	0.090***	0.074***	0.085***	0.485***	0.372***	0.442***	0.912***	0.675***	0.835***
	(0.013)	(0.024)	(0.011)	(0.069)	(0.120)	(0.059)	(0.130)	(0.237)	(0.112)
Postgraduate Degree	0.077***	0.082**	0.078***	0.412***	0.386**	0.393***	0.799***	0.722**	0.756***
	(0.019)	(0.033)	(0.016)	(0.093)	(0.153)	(0.079)	(0.176)	(0.289)	(0.149)
Financing - Savings	-0.020*	0.007	-0.012	-0.106*	0.055	-0.062	-0.203*	0.100	-0.116
	(0.012)	(0.021)	(0.010)	(0.058)	(0.103)	(0.050)	(0.106)	(0.201)	(0.091)
Financing - Bank Loan	0.078***	0.082***	0.081***	0.366***	0.435***	0.378***	0.691***	0.725***	0.702***
	(0.011)	(0.022)	(0.010)	(0.053)	(0.100)	(0.046)	(0.099)	(0.193)	(0.085)
Financing - Other Personal Assets	-0.003	-0.016	-0.002	-0.006	-0.109	-0.004	0.001	-0.222	0.009
	(0.011)	(0.023)	(0.010)	(0.060)	(0.124)	(0.053)	(0.113)	(0.245)	(0.099)

Financing - Credit Card	-0.040***	-0.065***	-0.045***	-0.255***	-0.397***	-0.271***	-0.458***	-0.751***	-0.506***
	(0.011)	(0.019)	(0.009)	(0.068)	(0.121)	(0.058)	(0.132)	(0.238)	(0.113)
Financing - Government Loan	0.002	0.121	0.025	0.065	0.472*	0.161	0.048	0.765	0.232
	(0.029)	(0.078)	(0.027)	(0.126)	(0.250)	(0.110)	(0.246)	(0.471)	(0.210)
Financing - Government Guaranteed Loan	0.154***	0.164***	0.154***	0.641***	0.702***	0.632***	1.138***	1.269***	1.124***
	(0.026)	(0.053)	(0.023)	(0.092)	(0.185)	(0.081)	(0.168)	(0.340)	(0.147)
Financing - Outside Investor	-0.000		0.0237*	0.012		0.138**	0.009		0.243**
	(0.020)		(0.013)	(0.101)		(0.0662)	(0.190)		(0.124)
Financing - None Needed	-0.051**		-0.043**	-0.433**		-0.377**	-0.850**		-0.748**
	(0.019)		(0.019)	(0.176)		(0.172)	(0.360)		(0.353)
Financing - Home Equity Loan		0.060***			0.339***			0.623***	
		(0.023)			(0.106)			(0.198)	
Financing - Loan from Family/Friends		0.002			0.022			0.051	
		(0.033)			(0.184)			(0.365)	
Financing - Other Sources of Capital		0.033			0.228			0.384	
		(0.044)			(0.216)			(0.413)	
Log(Population)	0.016**	0.000	0.012**	0.077*	0.002	0.056*	0.155**	0.022	0.113*
	(0.006)	(0.013)	(0.006)	(0.040)	(0.073)	(0.034)	(0.076)	(0.144)	(0.064)
Log(Personal Income)(Current $)	0.002	0.030	0.008	0.025	0.045	0.046	0.040	0.121	0.112
	(0.029)	(0.050)	(0.025)	(0.188)	(0.312)	(0.153)	(0.357)	(0.624)	(0.289)
Males (%)	-0.577	0.641	-0.370	-4.610	6.047	-2.909	-8.369	9.405	-5.469
	(0.512)	(0.909)	(0.441)	(3.290)	(5.372)	(2.851)	(6.349)	(10.60)	(5.467)
African Americans (%)	-0.048	0.174	-0.000	-0.395	1.224**	-0.090	-0.695	2.200*	-0.171
	(0.054)	(0.110)	(0.048)	(0.295)	(0.603)	(0.263)	(0.554)	(1.198)	(0.495)
Other Race (%)	-0.021	-0.193	-0.045	-0.175	-1.492	-0.295	-1.060	-2.658	-1.111
	(0.118)	(0.214)	(0.101)	(0.830)	(1.457)	(0.728)	(1.665)	(2.876)	(1.405)
Age: 14 and Younger (%)	0.668**	1.148**	0.739***	3.917**	7.962***	4.195***	7.521***	13.62**	8.087***
	(0.275)	(0.527)	(0.241)	(1.574)	(2.939)	(1.355)	(2.888)	(5.794)	(2.517)
Age: 65 and Older (%)	-0.022	0.442	0.048	-0.717	4.043	0.010	-1.442	7.010	-0.089
	(0.198)	(0.442)	(0.179)	(1.288)	(2.511)	(1.119)	(2.529)	(4.988)	(2.179)
At Least High School Degree (%)	-0.051	0.303	0.113	-0.706	1.439	0.235	-1.026	2.743	0.639
	(0.153)	(0.245)	(0.125)	(0.821)	(1.356)	(0.665)	(1.550)	(2.767)	(1.259)
Unemployed (%)	-0.733	-1.256	-0.383	-4.817	-13.48*	-3.197	-8.337	-25.71*	-5.223
	(0.491)	(1.108)	(0.402)	(3.015)	(7.429)	(2.400)	(5.850)	(15.15)	(4.576)
Log(Geographic Area)(Sq. Miles)	0.003	0.003	0.003	0.023	0.017	0.020	0.040	0.035	0.038
	(0.008)	(0.014)	(0.007)	(0.049)	(0.086)	(0.042)	(0.093)	(0.166)	(0.078)

MSA Central County (%)	-0.001	0.047	0.008	0.009	0.355*	0.064	-0.015	0.535	0.080
	(0.017)	(0.034)	(0.015)	(0.099)	(0.185)	(0.086)	(0.187)	(0.366)	(0.162)
MSA Outside County (%)	-0.014	-0.001	-0.012	-0.091	0.026	-0.065	-0.197	-0.019	-0.149
	(0.020)	(0.038)	(0.018)	(0.118)	(0.213)	(0.102)	(0.223)	(0.429)	(0.192)
FE - Industry (NAICS4)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - State	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - Year	No	No	Yes	No	No	Yes	No	No	Yes
Observations	5,240	1,690	6,930	5,240	1,690	6,930	5,240	1,690	6,930
R-squared	0.163	0.165	0.150						

Notes: Coefficients included in the regressions that cannot be disclosed due to U.S. Census Bureau restrictions are denoted by ND (Not Disclosed). For space reasons, some variables available and used in the 2007 regressions only and whose coefficients are subject to non-disclosure are not listed in the table. These are: "Financing - Grants, Venture Capital, Do Not Know", "Financing - Venture Capital", and "Financing - Do Not Know". Owner-level default categories are Female, Caucasian, Age: Younger than 35, and Less than High School and High School. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Robust standard errors in parentheses.

Table 5: The Effect of Franchising on Business Survival and Growth, Linear Probability Model

	I(Survival)			I(Marginal Survival)		Employment Growth	
VARIABLES	**1st Year**	**2nd Year**	**3rd Year**	**2nd Year**	**3rd Year**	**2nd Year**	**3rd Year**
Franchised Business Indicator	0.051***	0.064***	0.064***	0.021*	0.010	0.485***	-0.027
	(0.007)	(0.013)	(0.016)	(0.011)	(0.013)	(0.036)	(0.021)
Male	-0.005	-0.002	0.010	0.002	0.014	0.094***	0.019
	(0.007)	(0.011)	(0.012)	(0.0095)	(0.010)	(0.024)	(0.016)
African American	-0.058*	-0.106***	-0.131***	-0.068*	-0.057	-0.122	0.103*
	(0.031)	(0.040)	(0.043)	(0.037)	(0.041)	(0.084)	(0.060)
Asian	0.002	0.026*	0.045***	0.025**	0.025**	-0.302***	-0.022
	(0.009)	(0.014)	(0.016)	(0.012)	(0.013)	(0.033)	(0.022)
Other Race	ND	ND	ND	ND	ND	-0.118	0.010
	ND	ND	ND	ND	ND	(0.091)	(0.061)
Age: 35-44	0.001	0.013	0.021	0.013	0.012	0.030	0.027
	(0.008)	(0.012)	(0.014)	(0.010)	(0.012)	(0.027)	(0.018)
Age: 45-54	0.008	0.025**	0.030**	0.021*	0.011	0.067**	-0.023
	(0.009)	(0.013)	(0.015)	(0.011)	(0.012)	(0.029)	(0.019)
Age: 55-64	-0.014	-0.008	-0.017	0.005	-0.013	0.032	-0.017
	(0.012)	(0.018)	(0.020)	(0.015)	(0.017)	(0.039)	(0.025)
Age: 65 and Older	-0.001	0.042	0.049	0.053**	0.016	0.200**	-0.042
	(0.024)	(0.033)	(0.037)	(0.026)	(0.030)	(0.078)	(0.045)
Technical/Vocational School	0.003	0.048***	0.060***	0.051***	0.023	-0.011	-0.010
	(0.013)	(0.017)	(0.021)	(0.014)	(0.016)	(0.038)	(0.025)
Some College	-0.002	0.015	0.0025	0.021*	-0.014	0.067**	0.005
	(0.009)	(0.014)	(0.016)	(0.012)	(0.013)	(0.031)	(0.020)
Associate Degree	-0.003	0.003	-0.000	0.007	-0.004	0.037	0.029
	(0.014)	(0.020)	(0.023)	(0.017)	(0.018)	(0.047)	(0.028)
Bachelor's Degree	0.007	0.030**	0.027*	0.026**	-0.001	0.155***	-0.001
	(0.008)	(0.013)	(0.015)	(0.011)	(0.012)	(0.030)	(0.019)
Postgraduate Degree	-0.005	-0.002	-0.002	0.004	-0.004	0.121***	0.020
	(0.013)	(0.019)	(0.021)	(0.016)	(0.018)	(0.042)	(0.026)
Weekly Work: 20-39 Hrs.	0.012	0.032*	0.011	0.024	-0.019	-0.143***	0.056**
	(0.013)	(0.018)	(0.021)	(0.017)	(0.019)	(0.042)	(0.027)
Weekly Work: 40 Hrs.	0.035***	0.055***	0.078***	0.028*	0.039**	-0.143***	0.088***
	(0.012)	(0.019)	(0.021)	(0.017)	(0.017)	(0.043)	(0.027)

Weekly Work: 41-59 Hrs.	0.020*	0.061***	0.072***	0.049***	0.026*	-0.114***	0.028
	(0.011)	(0.016)	(0.018)	(0.014)	(0.015)	(0.036)	(0.023)
Weekly Work: 60+ Hrs.	0.014	0.042***	0.075***	0.035**	0.049***	-0.108***	0.070***
	(0.011)	(0.015)	(0.017)	(0.014)	(0.015)	(0.036)	(0.022)
Sole Proprietorship	-0.070***	-0.129***	-0.151***	-0.077***	-0.050***	-0.275***	-0.001
	(0.010)	(0.014)	(0.016)	(0.012)	(0.013)	(0.030)	(0.020)
Partnership	-0.029***	-0.057***	-0.083***	-0.033***	-0.042***	-0.406***	0.004
	(0.008)	(0.012)	(0.014)	(0.010)	(0.012)	(0.027)	(0.016)
Operated from Home	-0.081***	-0.112***	-0.115***	-0.054***	-0.024	-0.462***	-0.020
	(0.016)	(0.020)	(0.022)	(0.018)	(0.019)	(0.037)	(0.027)
Family Owned	0.003	0.002	0.009	-0.001	0.010	-0.010	0.018
	(0.006)	(0.010)	(0.011)	(0.008)	(0.009)	(0.023)	(0.013)
Initial Year Employment	-0.001	0.001*	0.003***	0.002***	0.002***	-0.059***	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.005)	(0.001)
Financing - Savings	0.004	0.010	0.014	0.007	0.008	-0.018	-0.003
	(0.007)	(0.011)	(0.013)	(0.010)	(0.011)	(0.026)	(0.016)
Financing - Bank Loan	0.019**	-0.002	-0.002	0.021**	0.011	0.226***	-0.020
	(0.007)	(0.012)	(0.014)	(0.009)	(0.010)	(0.025)	(0.015)
Financing - Other Sources	-0.014	-0.037***	-0.058***	-0.019*	0.000	0.056**	0.003
	(0.009)	(0.0123)	(0.014)	(0.010)	(0.011)	(0.026)	(0.017)
Financing - Credit Card	ND	ND	ND	-0.027**	-0.033***	0.004	-0.035**
	ND	ND	ND	(0.011)	(0.012)	(0.026)	(0.017)
Financing - Gov. Loan	-0.006	-0.019	-0.041	-0.017	-0.030	0.100*	0.026
	(0.015)	(0.024)	(0.029)	(0.021)	(0.025)	(0.057)	(0.035)
Financing - Gov. Guaranteed	0.029***	0.045***	0.047***	ND	ND	0.274***	-0.014
	(0.007)	(0.010)	(0.013)	ND	ND	(0.047)	(0.029)
Financing - Outside Investor	0.013	0.029*	0.025	0.021	0.002	0.089**	-0.023
	(0.010)	(0.015)	(0.018)	(0.013)	(0.014)	(0.037)	(0.023)
Financing - None Needed	0.014	-0.019	0.010	-0.033	0.036	-0.040	-0.054
	(0.021)	(0.031)	(0.034)	(0.028)	(0.026)	(0.068)	(0.037)
County-Level Market Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - Industry (NAICS4)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - State	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,930	6,930	6,930	6,420	5,690	6,420	5,690
R-squared	(0.007)	(0.013)	(0.016)	0.039	0.043	0.288	0.030

Notes: The sample is the 2002 and 2007 combined sample. Coefficients included in the regressions that cannot be disclosed due to U.S. Census Bureau restrictions are denoted by ND (Not Disclosed). Owner-level omitted categories are Female, Caucasian, Age: Younger than 25, and Less than High School. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Robust standard errors in parentheses.

Table 6: The Effect of Franchising on Business Survival and Growth with Propensity Score Matching and Sensitivity Tests

	I(Survival)			I(Marginal Survival)		Employment Growth	
VARIABLES	**1st Year**	**2nd Year**	**3rd Year**	**2nd Year**	**3rd Year**	**2nd Year**	**3rd Year**
Treated Sample Mean	0.979	0.892	0.797	0.914	0.894	1.822	-0.001
Control Sample Mean	0.922	0.831	0.722	0.900	0.874	1.287	0.059
Difference	0.057***	0.061***	0.075***	0.014	0.020	0.535***	-0.060**
S.E.	0.010	0.016	0.020	0.014	0.016	0.051	0.025
t-stats	5.59	3.77	3.71	0.99	1.25	10.41	-2.36
Rbounds Critical Cut-off (p = 0.05)	2.4	1.3	1.2	N/A	N/A	1.9	1
Maximum Allowable Deviations	(6.2%, 27.6%)	(10.9%, 17.1%)	(11.7%, 16.0%)	N/A	N/A	(7.7%, 23.1%)	N/A

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Rbounds critical cut-off is defined as the highest value of the bound (referred to as e^{γ} in Appendix B) such that the measured treatment effect is still statistically significant at the 5% level. Maximum allowable deviations are the largest deviations that an unobserved variable could maximally introduce without undermining the statistically significant results given that the average probability of starting a franchised business is 13.7%. The Rbounds critical cut-off values and the maximum allowable deviations are not available (N/A) if the original estimated results are statistically insignificant.

Table 7: The Effect of Franchising on Business Survival and Growth with Instrumental Variable Approach

	I(Survival)			I(Marginal Survival)		Employment Growth	
VARIABLES	**1st Year**	**2nd Year**	**3rd Year**	**2nd Year**	**3rd Year**	**2nd Year**	**3rd Year**
Franchised Business Indicator	0.021	-0.010	0.032	-0.024	0.045	1.142***	0.118
	(0.047)	(0.069)	(0.079)	(0.058)	(0.059)	(0.177)	(0.110)
Owner Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Business Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County-Level Market Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - Industry (NAICS4)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - State	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,930	6,930	6,930	6,420	5,690	6,420	5,690
R-squared	0.049	0.061	0.072	0.037	0.042	0.241	0.022

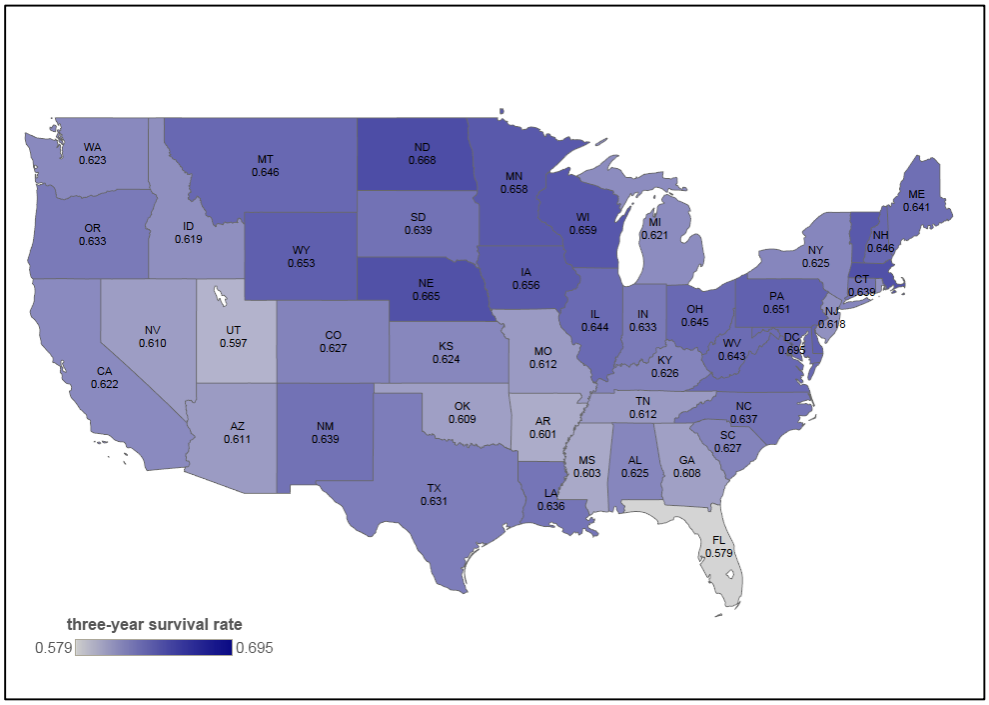
Notes: The sample is the 2002 and 2007 combined sample. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Robust standard errors in parentheses.

Table 8: The Effect of Franchising on Business Survival and Growth with Control Function Approach

	I(Survival)			I(Marginal Survival)		Employment Growth	
VARIABLES	**1st Year**	**2nd Year**	**3rd Year**	**2nd Year**	**3rd Year**	**2nd Year**	**3rd Year**
Franchised Business Indicator	-0.043	-0.169	-0.061	-0.125	0.109	2.218***	0.402
	(0.139)	(0.196)	(0.223)	(0.170)	(0.179)	(0.483)	(0.318)
Generalized Residual	0.059	0.147	0.079	0.092	-0.062	-1.090***	-0.270
	(0.087)	(0.123)	(0.140)	(0.107)	(0.112)	(0.303)	(0.199)
Owner Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Business Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County-Level Market Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - Industry (NAICS4)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - State	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE - Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,930	6,930	6,930	6,420	5,690	6,420	5,690
R-squared	0.051	0.065	0.073	0.039	0.043	0.290	0.031

Notes: The sample is the 2002 and 2007 combined sample. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Robust standard errors in parentheses.

Figure 1 Three-Year Survival Rates across States



Appendix A – Data

We focus on single-establishment businesses started in the years to which the Survey of Business Owners (SBO) pertains, namely, 2002 and 2007. We identify the starting and ending years of a business's existence by means of the Longitudinal Business Database (LBD), which tracks all businesses over time, and the SBO, which identifies when a business was originally established. Our sample is restricted to for profit, neither publicly held nor owned by another organization including trust, estate, or members of a club or cooperative, businesses organized as proprietorships, partnerships, or corporations with a startup year number of employees not exceeding one hundred. We also require that businesses not have been established for a one-time event and not be of a seasonal or occasional nature, and that the main owner holds at least a fifty percent share and works in the business. We exclude businesses owned by Alaska Native or American Indian tribal entities, businesses sold to another party by the time the survey was conducted, and businesses with missing or imputed values for the variables needed to apply the sample restrictions or conduct the analyses. SBO 2007 indicates whether a franchisor had more than a fifty percent stake in a business, and we exclude those for which this is true. Finally, as our focus is the retail, restaurant and services sectors, we restrict our analyses to sectors of NAICS2 of 44, 45, 72, and 81, and only industries (NAICS4) in which at least three percent of businesses in our cleaned SBO sample were franchised.

In terms of sample size, the SBO targeted 2,247,200 businesses in 2002 and 2,245,100 in 2007 and achieved response rates of 75 percent and 62 percent, respectively.[19] Focusing on single establishment businesses gives us 158,600 businesses in the 2002 and 96,700 businesses in the 2007 wave. Our data cleaning process, as noted above, has many steps. We list the most important ones from the perspective of sample size reduction. Considering only observations with non-

missing and non-imputed values, specifically in ownership share, leaves us with 98,300 businesses in the 2002 and 45,800 businesses in the 2007 wave.[20] Considering only businesses started in one of the years to which the SBO pertains, that is, only businesses that responded to the survey question, "When was the business originally established, purchased, or acquired by the owner(s) [..]?" with "2002" or "2007," reduces the 2002 sample to 32,200 businesses and 2007 sample to 18,300 businesses. Selecting only businesses that responded to the similar survey question, "In what year was this business originally established?" (missing in the 2002 SBO) with "2007" reduces the sample to 15,100 businesses. Considering only industries in which franchising was a relevant option for business owners reduced the 2002 sample to 15,400 and the 2007 sample to 6,000 businesses. Focusing only on retail, restaurant, and services sectors and making further cleaning steps result in final samples of 5,240 observations for the SBO 2002 wave and 1,690 observations for the SBO 2007 wave.

In the survival regressions, we control for the following business characteristics: legal form of organization (sole proprietorship, partnership, or corporation), industrial sector (4-digit NAICS), geography (state fixed effects), and startup number of employees. We also control for the following business owner characteristics: age (younger than 25, 25-34, 35-44, 45-54, 55-64, 65 and older), gender, race (White, African American, Asian, Other), education (less than high school, high school, technical or vocational school, some college, associate degree, bachelor's degree, postgraduate degree), and average weekly number of hours worked in the business (less than 20 hours, 20-39 hours, 40 hours, 41-59 hours, 60 hours or more). We include indicators for whether the majority of a business is owned by members of the same family and whether the business is operated primarily from a home. We include the following market conditions variables for the year in which a business was started: log of county population, log of county per capita

mean personal income, log of county geographic area, county fraction of males, county fraction of African Americans, county fraction of population other than African American or Caucasian, county fraction of population age 14 and younger, county fraction of population age 65 and older, and county unemployment rate. We also control for county fraction of population age 25 and older with at least a high school degree or equivalent, and include indicators for central and outlying counties of the Metropolitan Statistical Areas.

Appendix B – Rosenbaum Sensitivity Test

The primary assumption that underlies matching methods is the conditional independence assumption. In other words, the validity of a matching approach depends on the assumption that the treatment is a random assignment conditional on a set of observed variables, which can be denoted as

$$(\mathrm{Y}_0, \mathrm{Y}_1) \perp \mathrm{D}|\, \mathrm{X}.$$

The conditional independence assumption will be satisfied if X includes all of the variables that simultaneously influence both the franchising decision D and the outcome variables (survival and growth rate). Clearly, this is a strong assumption, which is hard to justify even if we have a very rich set of observed variables X. If there are unobserved variables that affect both the franchising decision and the outcome variables at the same time, the estimated average treatment effects suffer from "hidden bias" problem. Thus, it is important to check the sensitivity of the estimated results with respect to the potential "hidden bias."

Rosenbaum (2002) proposed a bounding approach to measure how the estimated average treatment effects change when an unobserved confounding factor is present in the treatment assignment. Assume two entrepreneurs with the exactly the same observed characteristics, i.e. $X_i = X_j$, then if the conditional independence assumption holds, we would expect the probability of starting a franchised business to be equal for the two entrepreneurs, i.e. $p(X_i) = p(X_j)$. However, if there is an unobserved confounding factor, then let us assume the following bounds on the odds-ratio that either of the two entrepreneurs will start a franchised business as:

$$\frac{1}{e^{\gamma}} \leq \frac{p(X_i)\big(1-p(X_j)\big)}{p(X_j)(1-p(X_i))} \leq e^{\gamma}.$$

If $e^{\gamma} = 1$, then we assume that there is no confounding factors that the two entrepreneurs who have the same observed characteristics would have the same probability of starting a

franchised business. If $e^{\gamma} = 2$, then we assume that the unobserved confounding factor can cause the odds-ratio of starting a franchised business with the two entrepreneurs who have the same observed characteristics to differ as much as a factor of 2. In this sense, e^{γ} is the bound for how much the unobserved confounding factor affects the treatment assignment. If the estimated results are still significantly significant given a large value of e^{γ}, then we conclude that the estimated average treatment effects using the matching approach is relatively insensitive to the "hidden bias." Otherwise, the estimated results are sensitive to the "hidden bias."

When the outcome variable is binary, we use the Mantel and Haenszel (MH) test statistic, which is implemented in Becker and Caliendo (2007).

Appendix C – Instrumental Variable Approach and Control Function Approach for the Binary Dependent Variable Case

In our application, as the endogenous variable is binary, we employ both the IV approach and CF approach with caution. Following Wooldridge (2010), for simplicity, we assume a linear probability model for binary outcomes and a linear model for continuous outcomes:

$$y_1 = \mathbf{z_1}\delta_1 + \alpha_1 y_2 + u_1,$$

where y_1 are the outcomes of interest, either the 1-year, 2-year, or 3-year business survival indicators or the growth rate in employment, y_2 is the possibly endogenous binary variable, here is the franchise business indicator variable, and $\boldsymbol{z_1}$ is a vector of exogenous explanatory variables. To estimate the effect of the franchising decision on business survival, we first estimate a probit model for y_2:

$$y_2 = I[\mathbf{z}\delta_2 + e_2 \geq 0],$$

where $\mathbf{z}$ is a vector including both z_1 and the instrumental variables, in this case, including the state-level proportions of franchises in the same industry and in related industries, and the state-level franchisor ownership rate among establishments of franchised chains in the same industry. From this estimation, we obtain the predicted probabilities $\Phi(\mathbf{z}\hat{\delta}_2)$, where $\hat{\delta}_2$ is the probit estimator.

In the IV approach, we use the predicted probabilities $\Phi(\mathbf{z}\hat{\delta}_2)$ as the instrumental variable and 2SLS to get consistent estimates.

In the CF approach, we use the predicted probabilities $\Phi(\mathbf{z}\hat{\delta}_2)$ to calculate the inverse Mills ratio (IMR): $\lambda(\cdot) = \phi(\cdot)/\Phi(\cdot)$ and the "generalized residual":

$$\widehat{gr}_2 \equiv y_2\lambda(\mathbf{z}\hat{\delta}_2) - (1 - y_2)\lambda(-\mathbf{z}\hat{\delta}_2).$$

In the second step, we use the "generalized residual" as an additional regressor in the following regression:

$$y_1 = z_1\delta_1 + \alpha_1 y_2 + \gamma_1 \hat{gr}_2 + e_1.$$

The consistency of the control function estimator hinges on the following assumptions: (1) $D(y_2|\boldsymbol{z})$ is correctly specified; (2) (u_1, e_2) is independent of z_1; and (3) $E(u_1|e_2) = \rho_1 e_2$, i.e. there exists a linear relationship between these two error terms.

Appendix D – Intermediate Results in IV and CF Approaches

Table D1. Summary Statistics for Instrumental Variables

	Franchised Businesses		Independent Businesses	
	Mean	S.D.	Mean	S.D.
State-Level Franchise Proportion in the Same Industry	0.38	0.30	0.18	0.21
State-Level Franchise Proportion in the Related Industries	0.27	0.18	0.21	0.19
State-Level Franchisor Ownership Rate among Establishments of Franchise Chains in the Same Industry	0.18	0.17	0.20	0.23
Number of Observations	950		5980	

Table D2. The First Step in IV and CF Approach

VARIABLES	Probit Model Franchise Indicator
State-Level Franchise Proportion in the Same Industry	1.328***
	(0.164)
State-Level Franchise Proportion in Related Industries	0.478
	(0.383)
State-Level Franchisor Ownership Rate among Establishments of Franchise Chains in the Same Industry	-0.428***
	(0.128)
Business Owner Characteristics	Yes
Business Characteristics	Yes
Market Level Characteristics	Yes
FE - Industry	Yes
FE - State	Yes
FE - Year	Yes
Observations	6,930

Table D3. The First Stage regression in IV Approach

VARIABLES	Franchise Indicator
Predicted Probability of Franchising	1.065***
	(0.074)
Business Owner Characteristics	Yes
Business Characteristics	Yes
Market-Level Characteristics	Yes
FE - Industry	Yes
FE - State	Yes
FE - Year	Yes
Observations	6,930
R-Squared	0.195
Cragg-Donald Wald F statistic	301.86

Note: The Stock-Yogo weak identification test critical value for this regression is 16.38. As the estimated Cragg-Donald Wald F statistic is greater than this critical value, we reject the null hypothesis of weak identification at the standard 10% level. The reported results are for the first stage in the unconditional survival analysis. The results of conditional survival analysis and the growth analysis are similar to the above.

[1] See Blair and Lafontaine (2005), Chapter 2, for an overview of this literature.

[2] The author attributes these high failure rates to the saturation and diminishing returns for newcomers in popular franchising niches.

[3] A related but separate literature considers factors that affect the decision to start a business. See notably Hurst and Lusardi (2004), Holtz-Eakin et al. (1994), Blanchflower and Oswald (1998), Kerr and Nanda (2011), and Franco (2005).

[4] NAICS stands for North American Industry Classification System, which is the standard used by Federal statistical agencies.

[5] The Characteristics of Business Owners (CBO) database used by Bates (1995a, b, 1998) and Williams (1999) was the precursor of the SBO. The CBO sample was much smaller, however.

[6] More information about the LBD can be found in Jarmin and Miranda (2002).

[7] Specifically, the concern is that as the proportion of franchise relative to independent businesses that survive for any period of time may, as data below suggest, be different, such that conclusions based on subsets of businesses that have survived up to the survey year could bias our main coefficient of interest.

[8] To satisfy U.S. Census Bureau disclosure procedures, counts in this paper are rounded to hundreds or tens.

[9] See Appendix A for a detailed description of the data and how we arrive at our final sample.

[10] If one believes that business sales should not be counted as business failures, how this would affect our results depends on whether the buying and selling of franchises relative to independent businesses exhibits systematic differences. Assuming a similar pattern for both types of businesses, our descriptive statistics would underestimate to a similar degree the true survival rates of both types of businesses. Thus, the difference in survival rates between the two types of businesses

would be correctly estimated, and the coefficient on our franchise indicator variable in regression analyses would be unbiased. If we expect franchises to be more likely than independent businesses to be bought and sold, if for no other reason than that the former are less likely to be substantially changed upon transfer of ownership, then the difference in survival rates, and coefficient on franchise status in our regressions, would be biased downwards. In other words, our results below would be conservative under such assumptions.

[11] There are two potential reasons for the small employment numbers in the first year. First, this may be because a business starts relatively small. Second, if a business starts in the middle of a year, the number of employees as captured in the LBD on a single date in March may underestimate the full year level of employment. Since the issue of the small employment numbers in the first year affects both franchised and independent businesses in similar ways, the results comparing the effect of the two business formats on business growth rates should be unaffected.

[12] Note that government guaranteed loans are different from government loans. In the latter case, the government does not provide a loan but instead provides guarantees to banks or financial institutions, thereby protecting them against defaults on payments. This alleviates the risk associated with making these loans. SBA-guaranteed loans are a type of government-guaranteed business loan.

[13] We performed all the growth analyses using both growth in employment and growth in payroll. As the results from the two measures were very similar, for space reasons, we only report findings from the employment growth regressions.

[14] The family owned variable is coded based on a question that asks whether the majority of the business is owned by members of the same family.

[15] We did not include effort level and other business owner decisions in our analyses in Table 4 as these are chosen along with, or even after, the decision to open a franchise is made. However, these variables are controlled for in our survival and growth analysis.

[16] For more details, see Appendices B and C.

[17] In other words, our probit estimations rely on the larger set of explanatory variables in Table 5, minus of course the franchise indicator variable.

[18] If a 6-digit NAICS level measure is unavailable due to too few observations, we use the 4-digit NAICS level instead. Specifically, we calculate the state-level proportion of businesses of franchised chains in the same industry (4-digit NAICS level) and in related industries (different 4-digit but same 2-digit NAICS code), and the state-level proportion of company-owned establishments among establishments of franchised chains in the same 4-digit NAICS industry to impute values for these variables when the 6-digit NAICS level measures are missing.

[19] To satisfy U.S. Census Bureau disclosure procedures, all counts are rounded to hundreds.

[20] Considering non-missing and non-imputed values for other key variables would result in similar final sample sizes.